

	Biparental Cross	GWAS	MPP
Allelic Diversity	Two alleles per locus, limited to variation between the two parents	High; captures population-wide standing variation	Scales with founder number (typically 4–25+); higher than biparental, more controlled than GWAS
Mapping Resolution	Depends on generations of recombination; limited in F ₂ , improves with further intercrossing or RIL generations	High; reflects recombination accumulated over many generations in a natural population	Improves with generations of intercrossing; fully intercrossed designs accumulate recombination faster per generation than hub-spoke structure because of intercrossing all founders vs pairs
Experimental Control	High; cross and environment fully controlled	Low; natural population, confounded by population structure and environment	High; cross and environment fully controlled, as in biparental designs
Sample Size	Modest; power comes from large allele frequency differences between two known parents	Very large (thousands to hundreds of thousands) to detect small effect sizes	Moderate; smaller than GWAS, especially with bulk segregant analysis
Founder Allele Representation	Equal by design (50/50)	Not founder-based; reflects natural population allele frequencies	Hub and spoke: unequal, spoke alleles present in $\sim 1/(n-1)$ of RILs. Fully intercrossed: approximately equal across founders
Maintenance Burden	RILs require ongoing line maintenance unless using an F ₂ /bulk segregant approach	None; uses existing natural population, no lines maintained	RIL panels require maintenance; bulk segregant approaches avoid this by pooling directly